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Quadrilaterals



Making video games for computers involves a lot of geometry. Sprite characters like Super Mario are composed of hundreds of small coloured squares. A popular game called "Minecraft" is totally based on using quadrilaterals to create exciting new virtual worlds.



What's Inside

Topic(s)

- ✓ Topic-wise Notes
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Questions

- ✓ New Pattern Qs
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- ✓ Detailed Explanation

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TOPIC 1

*INTRODUCTION TO QUADRILATERALS

Learning Objectives

- Students will be learning the concept of quadrilaterals.
- Students will be learning the angle sum property of a quadrilateral.
- Students will be learning the different types of quadrilaterals and their properties.

Learning Outcomes

- Students will be able to identify the sides, angles and vertices of a quadrilateral.
- Students will be able to determine the angles of a quadrilateral.
- Students will be able to determine the similarities and differences between quadrilaterals by looking at their sides, angles, and diagonal measures.

Real Life Application

You must have seen many buildings in your locality that are quadrilateral in shapes. Quadrilaterals are one of the most popular shapes used in making of buildings and homes. Have you ever thought about why architects use quadrilaterals to build beautiful buildings?

Quadrilaterals can be used to create multiple shapes. Quadrilaterals can be used to create rooms, the walls of all dwellings, and the floor in most cases. The architectural design of certain houses, for example, may be based solely on the utilisation of diverse quadrilateral shapes. There are other irregular quadrilaterals that have the capacity to make things look good. Because of their inability to be uniform, they make good designs and attract the eye. Hence, they are used mostly by architects.

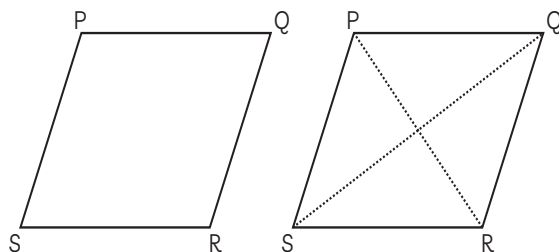


* General information about quadrilaterals are given only for knowledge purposes. Only axioms are there in the syllabus.

Quadrilaterals are preferred in constructing buildings over other polygons as they can make maximum use of space.

Let us now discuss quadrilaterals in detail.

A quadrilateral is a closed figure which has four sides, four angles, and also four vertices (as shown in the figure below).



In above figure,

PQRS is a quadrilateral

Sides of quadrilateral $\rightarrow$ PQ, QR, RS, SP

Vertices of quadrilateral $\rightarrow$ P, Q, R, S

Angles of quadrilateral $\rightarrow$ $\angle P, \angle Q, \angle R, \angle S$

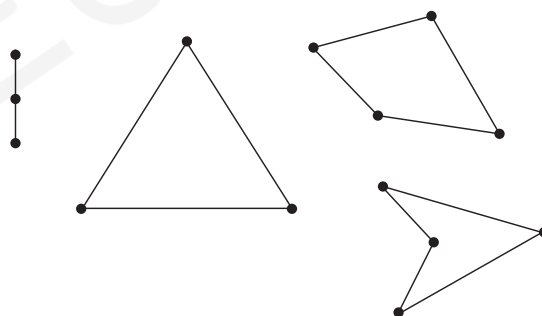
Now, join the opposite vertices of quadrilateral P, Q, R and S in the figure to get its diagonals.

Diagonals of quadrilateral PQRS are PR and QS.



Important

$\rightarrow$ If we join three collinear points, we get a line segment but if we join three non-collinear points, we get a shape of triangle. If we join four points among which any three points are non-collinear, we get a shape of quadrilateral.



Caution

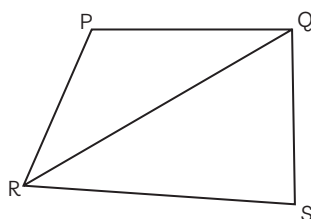
$\rightarrow$ A closed figure with n number of the sides is known as a polygon.

$\rightarrow$ Triangle is a polygon with least number of sides. The quadrilateral is also a polygon with 4 sides.

Angle Sum Property of a Quadrilateral

The sum of angles of a quadrilateral is 360° because it is made up of two triangles PQR and QRS as shown in figure below.





We know that,

In ΔPQR ,

$$\angle PQR + \angle QPR + \angle PRQ = 180^\circ$$

..... (i)

Similarly, In ΔQRS ,

$$\angle RQS + \angle QSR + \angle SRQ = 180^\circ$$

...(ii)

Adding eq. (i) and (ii), we get

$$\begin{aligned} \angle PQR + \angle QPR + \angle PRQ + \angle RQS + \angle QSR + \angle SRQ \\ = 180^\circ + 180^\circ \end{aligned}$$

$$\text{Also } \angle RQS + \angle PQR = \angle Q$$

$$\text{And } \angle PRQ + \angle SRQ = \angle R$$

$$\therefore \angle P + \angle Q + \angle R + \angle S = 360^\circ$$

i.e., The sum of the four angles of a quadrilateral is 360° .

Important

→ Sum of the angles of a polygon = $(n - 2) \times 180^\circ$

Where n = number of sides

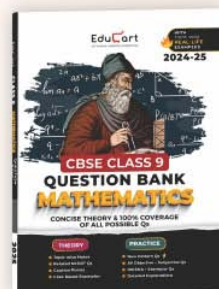
Now, in a quadrilateral, $n = 4$

$$\begin{aligned} \text{Sum of angles of quadrilateral} &= (4 - 2) \times 180^\circ \\ &= 2 \times 180^\circ \\ &= 360^\circ \end{aligned}$$



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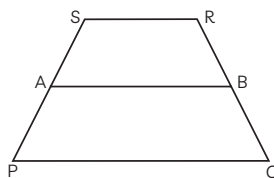


OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. PQRS is a trapezium in which $PQ \parallel SR$, $SR = 40$ cm and $PQ = 60$ cm. If A and B are mid-points of PS and QR, then what will be the value of $\frac{\text{ar}(\text{SABR})}{\text{ar}(\text{APQB})}$?

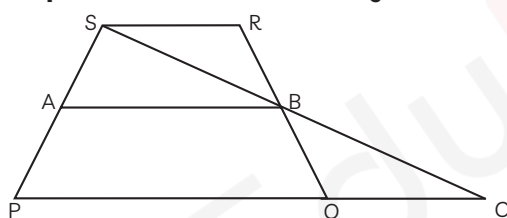


- (a) $\frac{9}{7}$ (b) $\frac{11}{9}$
(c) $\frac{7}{9}$ (d) $\frac{9}{11}$

[Diksha]

Ans.(d) $\frac{9}{11}$

Explanation: Extend PQ and SB and join them. In $\triangle SRB$ and $\triangle CQB$,



$$\angle SRB = \angle CQB$$

[Alternate interior angles]

$$RB = QB$$

[Q is the mid-point of RQ]

$$\angle SBR = \angle CBQ$$

[Vertically opposite angles]

$$\therefore \triangle SRB \cong \triangle CQB$$

[By ASA Congruence Rule]

$$\therefore SB = CB$$

[By CPCT]

Hence, we can say that B is the mid-point of SC.(i)

$$SR = CQ$$

[By CPCT]

$$CQ = 40 \text{ cm [SR = 40 cm]}$$

...(ii)

$$PC = PQ + QC$$

$$PC = 60 + 40$$

[Given from eq. (ii)]

$$\therefore PC = 100 \text{ cm}$$

...(iii)



In $\triangle PSC$, point A is the mid-point of side SP and given point B is the mid-point of side SC.

[From eq. (i)]

$\therefore$ By mid-point theorem,

$$AB \parallel PC \text{ and } AB = \frac{1}{2} PC$$

$$\therefore AB = \frac{1}{2} \times 100$$

[From eq. (iii)]

$$\therefore AB = 50 \text{ cm}$$

Area of a trapezium

$$= \frac{(\text{sum of lengths of parallel sides})}{2} \times \text{height}$$

$$\text{Area of SABR} = \frac{(SR + AB)}{2} \times H$$

$$\text{Area} = \frac{(40 + 50)}{2} \times H$$

$$\text{Area} = \frac{90}{2} \times H$$

$$\text{Area} = 45H$$

$$\text{Area of APQB} = \frac{(AB + PQ)}{2} \times H$$

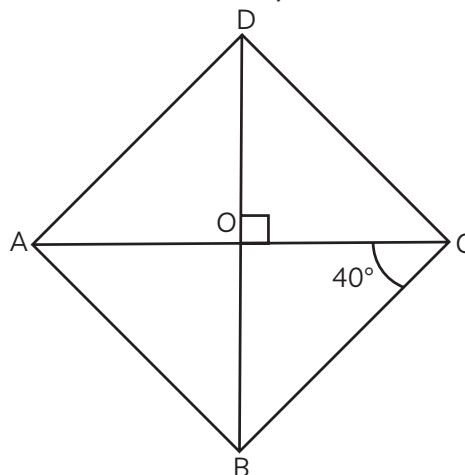
$$= \frac{(50 + 60)}{2} \times H$$

$$= \frac{110}{2} \times H$$

$$\text{Area} = 55H$$

$$\frac{\text{ar}(\text{SABR})}{\text{ar}(\text{APQB})} = \frac{45H}{55H} = \frac{9}{11}$$

2. ABCD is a rhombus. If $\angle ACB = 40^\circ$, then $\angle ADB$ is :



- (a) 90° (b) 80°
 (c) 60° (d) 50°

[Delhi Gov. QB 2022]

Ans.(d) 50°

Explanation: In given figure, ABCD is a rhombus. We know that diagonals of rhombus bisect each other perpendicularly.

Hence, $\angle BOC = 90^\circ$

$\angle OCB = 40^\circ$

[Given]

$AD \parallel BC$ and BD is the transversal

[Opposite sides of rhombus are parallel to each other]

$\therefore \angle ADB = \angle DBC$

[Alternate angles]

In $\triangle OBC$,

$\angle BOC + \angle OCB + \angle OBC = 180^\circ$

$\Rightarrow 90^\circ + 40^\circ + \angle OBC = 180^\circ$

$\Rightarrow \angle OBC = 180^\circ - 130^\circ$

$\therefore \angle OBC = 50^\circ$

But $\angle OBC = \angle DBC$

$\therefore \angle DBC = 50^\circ$

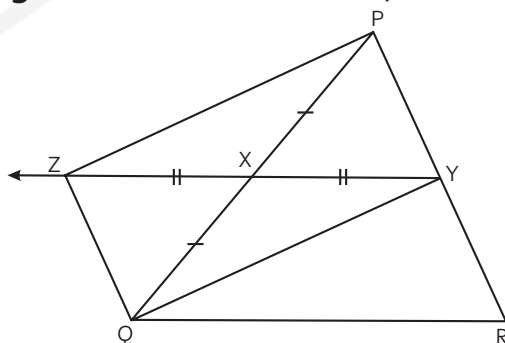
But $\angle ADB = \angle DBC$

$\therefore \angle ADB = 50^\circ$

[Alternate angle]

True and False

3. Points X and Y are the mid-points of PQ and PR of $\triangle PQR$ respectively. If point Z is on ray XY such that $YX = XZ$, then PZQY is a rectangle.



[Diksha]

Ans. False

Explanation: Points X and Y are the mid-points of PQ and PR of $\triangle PQR$ respectively. Point Z is on ray XY such that $YX = XZ$.
 [Given]



PQ and YZ are the diagonals of PZQY.

$$PX = XQ$$

[Point X is the mid-point of PQ]

$$XZ = XY$$

[Given]

$\therefore$ Diagonals of PZQY are bisecting each other.

We can say that PZQY is a parallelogram because if the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

Assertion and Reason (A-R)

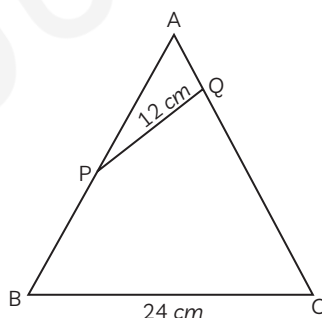
In these questions, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

4. Assertion (A): In a triangle ABC, two points P and Q are on the sides AB and AC respectively such that $PQ = 12$ cm. If the length of BC is 24 cm, then PQ must be parallel to BC.

Reason (R): The line segment joining the mid-points of two sides of a triangle is parallel to the third side.



Ans. (d) Assertion (A) is false but reason (R) is true.

Explanation: PQ is parallel to BC only if P and Q are mid-points of AB and AC respectively.

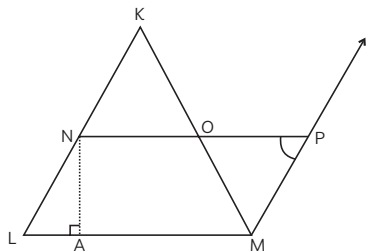
CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

5. Ananya wants to frame a question based on quadrilaterals. So, she draws a triangle KLM (as shown in figure below) where N and O are mid-points of side KL and KM of $\triangle KLM$.

Now, she wants to prove $NO \parallel LM$. She constructs a ray l such that $KL \parallel MP$



- (A) Ananya proved $\triangle KON \cong \triangle MOP$ by using congruence rule.
- (a) SAS (b) ASA
(c) SAA (d) AA
- (B) If $LM = 12$ cm, then half of NO is:
- (a) 6 cm (b) 10 cm
(c) 8 cm (d) 3 cm
- (C) If $\triangle KON \cong \triangle MOP$, then which of the following statements is false?
- (a) $KO = OM$ (b) $\angle KON = \angle MOP$
(c) $\angle NKO = \angle MPO$ (d) $KN = MP$
- (D) NLMP is a
- (a) trapezium (b) parallelogram
(c) rectangle (d) rhombus
- (E) The area of NOML, if $NA \perp LM$ and $NA = 7$ cm, $NO = 6$ cm and $LM = 12$ cm is:
- (a) 65 cm^2 (b) 63 cm^2
(c) 60 cm^2 (d) 36 cm^2

Ans. (A) (b) ASA

Explanation: In $\triangle KON$ and $\triangle MOP$,

$$KO = OM$$

[O is mid-point]

$$\angle KON = \angle MOP$$

[Vertically opposite angles]

$$\angle NKO = \angle PMO$$

[Alternate angles]

Therefore, $\triangle KON \cong \triangle MOP$

[By ASA Congruence Rule]

(B)(d) 3 cm

Explanation: According to the question,



N and O are the mid-points

So, By mid-point theorem

$$NO = \frac{1}{2} LM$$

$$NO = \frac{1}{2} \times 12 \text{ cm} \\ = 6 \text{ cm}$$

$$\text{So, } \frac{1}{2} NO = \frac{1}{2} \times 6 \text{ cm} \\ = 3 \text{ cm}$$

(C)(c) $\angle NKO = \angle MPO$

Explanation: $\angle NKO = \angle PMO \neq \angle MPO$

[$\triangle KON \cong \triangle MOP$]

(D)(b) *parallelogram*

Explanation: As according converse of mid-point theorem, $NO = OP$ and $NL = PM$

Therefore, NLMP is a parallelogram.

(E)(b) 63 cm^2

Explanation: NOML is a trapezium, because $NO \parallel LM$ but NL is not parallel to OM.

Area of Trapezium

$$= \frac{1}{2} \times \text{height} \times [\text{sum of parallel sides}] \\ = \frac{1}{2} \times 7 \times [12 + 6] \\ = \frac{1}{2} \times 7 \times 18 \text{ cm}^2 \\ = 63 \text{ cm}^2$$

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

6. Three angles of a quadrilateral are 75° , 90° and 75° . Find the measure of the fourth angle. [Delhi Gov. QB 2022]

Ans.Given: Three angles of a quadrilateral are 75° , 90° and 75° .

Let the fourth angle be x.

$$75^\circ + 90^\circ + 75^\circ + x = 360^\circ$$

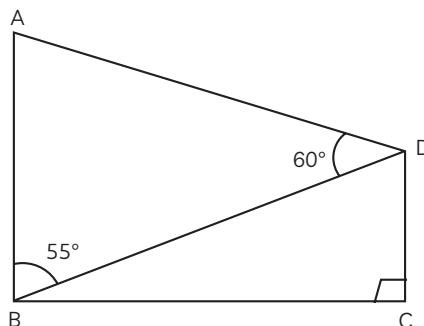
$$\Rightarrow 240^\circ + x = 360^\circ$$

$$\Rightarrow x = 360^\circ - 240^\circ$$

$$\Rightarrow x = 120^\circ$$

So, the measure of the fourth angle is 120° .

7. In the figure, $AB \parallel CD$, what will be the measure of $\angle ADC$?



[Delhi Gov. QB 2022]

Ans. $AB \parallel CD$

BD is transversal

$$\angle ABD = \angle BDC = 55^\circ$$

[Alternate angles]

$$\angle ADC = \angle ADB + \angle BDC$$

$$\angle ADC = 60^\circ + 55^\circ$$

$$\Rightarrow \angle ADC = 115^\circ$$

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

8. The angles of a quadrilateral are $(x + 20)$, $(x - 20)$, $(2x + 5)$, $(2x - 5)$. Find the value of x .
[Delhi Gov. QB 2022]

Ans. Sum of all angles in a quadrilateral is 360° .

$$(x + 20) + (x - 20) + (2x + 5) + (2x - 5) = 360^\circ$$

$$2x + 4x = 360^\circ$$

$$6x = 360^\circ$$

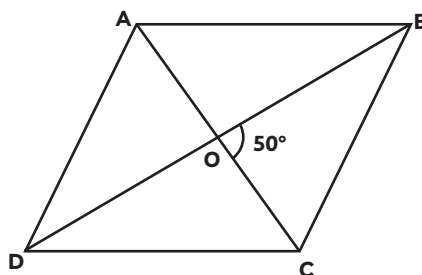
$$x = 60^\circ$$

SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

9. The diagonals of a rectangle ABCD meet at O. If $\angle BOC = 50^\circ$, then find $\angle ODA$.





[Delhi Gov. QB 2022]

Ans. $\angle BOC = \angle AOD$

(Vertically opposite angles)

$$\angle AOD = 50^\circ$$

In $\triangle AOD$, Since, the diagonals are equal, thus the bisectors will also be equal.

Thus, $OA = OD$

$$\therefore \angle OAD = \angle ODA$$

$$= \frac{1}{2}(180^\circ - 50^\circ)$$

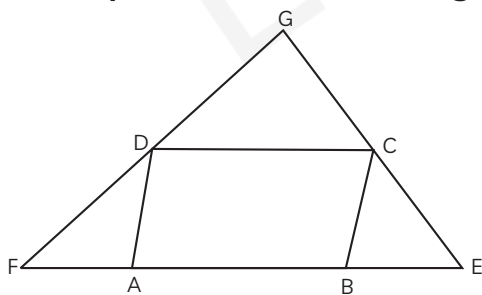
$$= \frac{1}{2}(130^\circ)$$

$$= 65^\circ$$

LONG ANSWER Type Questions (LA)

[4 & 5 marks]

10. ABCD is a parallelogram. Side AB is produced on both sides of E and F as in figure such that $BE = BC$ and $AF = AD$. Show that EC and FD when produced meets at right angle.



[Delhi Gov. QB 2022]

Ans. $\because BE = BC$ and $AF = AD$

$\therefore \triangle ADF$ and $\triangle BCE$ are isosceles triangle.

In $\triangle ADF$

$$\angle ADF = \angle AFD = \angle GDC$$

$$[\because DC \parallel AB \Rightarrow DC \parallel EF]$$

$$[\because \triangle ADF \text{ is an isosceles } \Delta]$$

In $\triangle ADC$,

$$\begin{aligned}\angle ADC &= \angle GDF - \angle GDC - \angle ADF \\ &= 180^\circ - \angle GDC - \angle ADF \\ &= 180^\circ - 2\angle AFD\end{aligned}$$

$$\angle ABC = 180^\circ - 2\angle AFD$$

($\because \angle ADC = \angle ABC$ i.e., opposite angles of parallelogram)
 $\Rightarrow \angle EBC = 2\angle AFD$

[$\angle ABE$ is straight line = 180°]

In $\triangle BCE$

$$\angle CBE + \angle BCE + \angle BEC = 180^\circ$$

[Angle sum property]

$$2\angle AFD + 2\angle BEC = 180^\circ$$

$$\angle AFD + \angle BEC = 90^\circ$$

In $\triangle GFE$

$$\angle G + \angle E + \angle F = 180^\circ$$

[By angle sum property]

$$\angle G + 90^\circ = 180^\circ$$

$$\angle G = 180^\circ - 90^\circ = 90^\circ$$

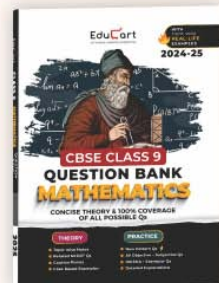
$$\angle FGE = 90^\circ$$

Hence, proved.



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